

Project title: Detection of helicobacter pylori in gastric biopsy

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1. Introduction

Problem Statement

Helicobacter pylori (*H. pylori*) is a tiny, spiral-shaped bacteria that lives in the stomach and affects more than half of the world's population. This bacterium can cause serious health problems—including stomach ulcers, chronic inflammation, and stomach cancer. Health organizations, including the World Health Organization, have officially classified *H. pylori* as a cancer-causing agent, making it a significant global health concern.

Why This Matters

Catching *H. pylori* early can prevent these serious health complications. With the right treatment, doctors can eliminate the bacteria before it causes lasting damage. But here's the catch: detecting it isn't easy. Right now, doctors examine stomach tissue samples under a microscope—a process that's slow, tiring, and inconsistent. Because *H. pylori* is incredibly small and can look different in different samples, even trained specialists sometimes miss it. While alternative methods such as immunohistochemistry offer improved accuracy, their high cost and technical demands limit their widespread use.

Proposed Solution

Our project aims to make *H. pylori* detection faster, more reliable, and accessible to everyone. By leveraging modern technology and digital tools, we want to help doctors spot this bacteria more consistently in stomach tissue samples. The goal is simple: better detection means earlier treatment. We're working to make this critical health screening available to more people, everywhere.

2. Literature review

The primary challenge our project addresses is the detection of tiny, few-pixel long *H. pylori* bacteria within massive, gigapixel sized Whole Slide Images (WSIs). This "needle in a haystack" problem makes manual checks by pathologists a time-consuming and labor-intensive task. Based on the research by Klein et al. we can conclude that a "coarse-to-fine" pipeline is the most practical approach. The researchers first used standard image processing to find "hot spot" regions (gastric glands) and then applied a CNN classifier to patches from these regions. This two-stage idea is impressive, but it's simple method for finding glands is a major bottleneck of the research.

This limitation was evident in our Seniors' research paper, which adopted a similar strategy. Their "template matching" technique for finding candidate glands was unreliable, missing almost 20% of the true positive regions before the AI model even had a chance to analyze them. This proves that the first step of accurately identifying all the relevant gland regions requires a more intelligent and robust solution, such as the U-Net segmentation model we propose.

The seniors' work also explored a more advanced object detection pipeline, similar to our current plan, using NuClick to mask/define necessary glands and a YOLO model/CNN model to find the bacteria. However, their results were constrained by a different, more fundamental problem

of poor annotations. Their paper explicitly states their ground-truth labels were "not entirely precise" and "limited" with many true positives likely mislabeled as negative. This unreliable data makes it extremely difficult to train a high-performance detector, which is reflected in their final results.

This "imprecise label" problem is a known challenge in digital pathology. Yang et al. tackled this issue by pioneering a "weakly supervised multi-task learning" framework. They recognized that perfect, pixel-level labels are impractical to obtain. Their solution was to train a single model using two imperfect targets simultaneously: one "broad" target (where false positives are possible) and one "fine-grained" target (where false negatives might show up). This forces the model to find a robust compromise. Their work shows a clear path forward, suggesting that our project can succeed by adopting similar weak-supervision techniques to overcome the exact annotation challenges that limited previous efforts if we fail to obtain the precise annotations after our classification/detection and re-annotation(by Philipp) loop.

3. Research Question, Hypothesis or Aim and Objectives

- Handle massive image files efficiently – Gastric biopsy slides are with billions of pixels, creating significant computational challenges that we need to solve.
- Build an automated detection system that can identify H. pylori in stomach tissue samples quickly and accurately.
- Speed up diagnosis by creating a faster alternative to the current manual process, helping doctors get results sooner.
- Make detection affordable and accessible so that hospitals with limited resources can also benefit from accurate H. pylori screening.
- Enable early treatment by catching infections before they lead to serious complications like ulcers or cancer.

4. Research design/methodology

The proposed approach involves object detection method in a hierarchical fashion.

- Pre-processing and BG filtration First we start with cleaning the data, by removing the unwanted pixels from the image, mainly white spaces. For this, we are proposing a HSV(Hue, Saturation, Value) color space approach which can detect lightning variations in biopsy slides (OTSU approach). WSI patches are converted to HSV color space, and then background regions are saturated, which removes 60-70% irrelevant data.
- Next, we are proposing the QuPath Algorithm, in which we can segment the gastric gland(most presence of h.pylori bacteria). The built in pixel classifier can reproduce the same work of NuClick in a faster way. In case this approach fails, the NuClick/U-Net model can be adapted.
- Sliding Window Extraction In case the segmented image is of high resolution, the sliding window can be considered. The image is divided into 320x320 pixel patches, with a 20% proposed overlap. This size is selected as its good for providing enough context to the model for recognizing tissue structures.

- From our analysis, we observe that the dataset has missing annotations. We propose to use a active learning approach, where the baseline model(preferable YOLO) can be trained using available annotated positive and negative data, and then it can be used to annotate the other samples in our main dataset
- Object Detection Approach For the segmented image, the YOLO classifier can be used, which can return the local positions of the bacteria present in the images.
- Our strategy to address the challenging 1 : 100 class imbalance combines three powerful techniques: implementing Focal Loss for robust minority-class recognition, employing aggressive data augmentation (geometric and color jitter) to simulate H&E stain variability, and utilizing batch-aware oversampling to ensure a minimum number of positive (bacteria)examples are present in every training batch.
- To achieve the final output of bounding boxes on the original full WSI, we use the patch's global anchor coordinates (top-left corner relative to the WSI) to transform the YOLO model's local coordinates (within the patch) into the final global coordinates for each detection.

5. Project plan

<https://github.com/users/hxrshx/projects/1/views/4?visibleFields=>

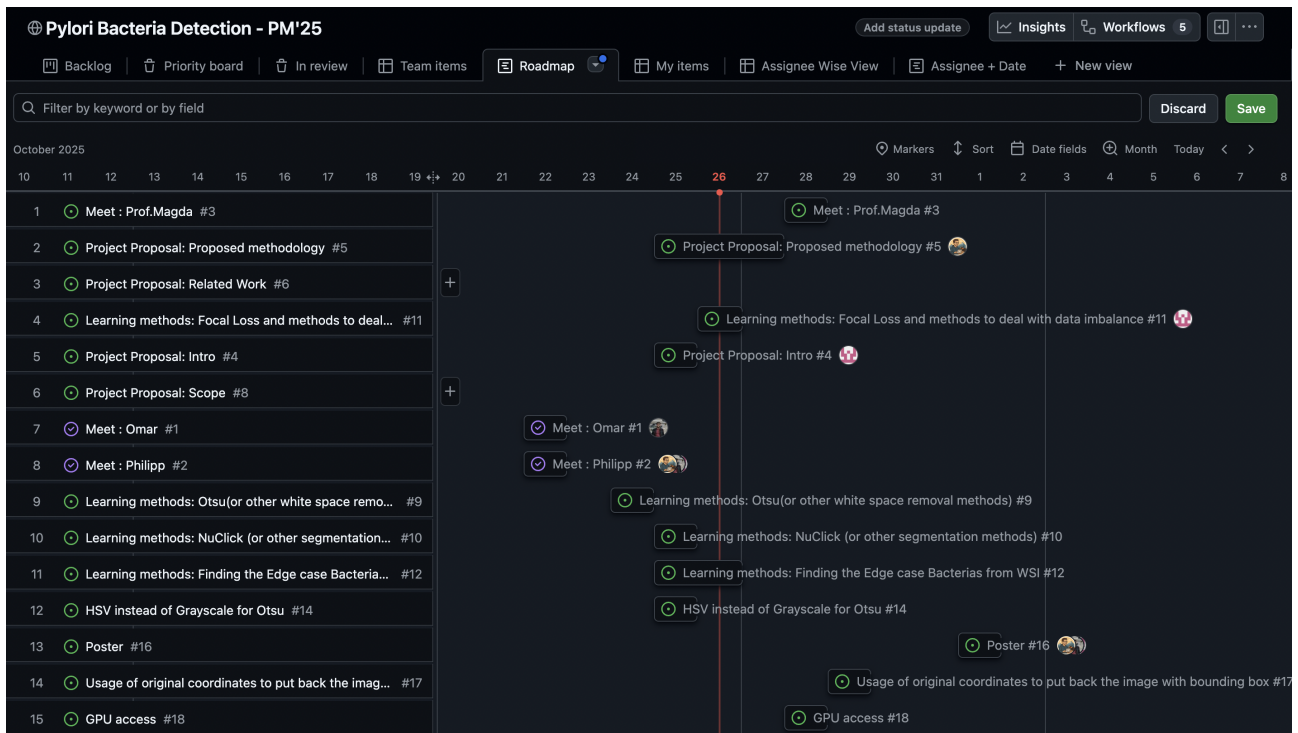


Figure 1: Current Task based Project Plan

6. Expected Outcome/ Implications and Contribution to the knowledge

The end result will be a solution for pathologists, available in a simple web interface. The pathologist uploads a Whole Slide Image (WSI) and our system analyses the WSI to identify

and rank the 20-30 most likely areas for *H. pylori*. The results will be presented as color-coded bounding boxes on the WSI: 10-15 red boxes for the highest confidence "hot spots" and 10-15 yellow boxes for the next highest confidence. This format fundamentally changes the review such that the pathologist is able to identify and concentrate on the most crucial situations immediately (red boxes, yellow boxes) in order to quickly confirm a positive case or confidently clear a slide as negative.

References

- Klein, Sebastian, Jacob Gildenblat, Michaela Angelika Ihle, Sabine Merkelbach-Bruse, Ka-Won Noh, Martin Peifer, Alexander Quaas, and Reinhard Büttner (2020). "Deep learning for sensitive detection of *Helicobacter Pylori* in gastric biopsies". en. In: *BMC Gastroenterology* 20.1, p. 417.
- Yang, Yongquan, Yiming Yang, Yong Yuan, Jiayi Zheng, and Zheng Zhongxi (2020). "Detecting helicobacter pylori in whole slide images via weakly supervised multi-task learning". en. In: *Multimedia Tools and Applications* 79.35-36, pp. 26787–26815.